BTS-III(15)-11.17-1557 Reg.No. [ ][ ][ ][ ][ ][ ][ ][ ]

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B.Tech. Degree III Semester Examination November 2017

EC 15-1302 DISCRETE COMPUTATIONAL STRUCTURES
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

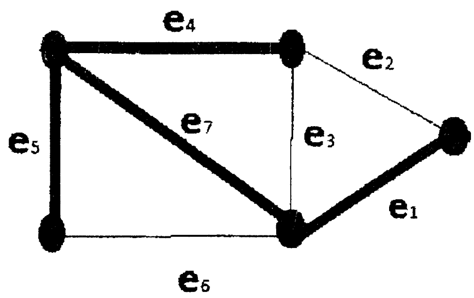
PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Explain the different closure properties of relations.

(b) Let $X = \{1, 2, 3\}$, $Y = \{a, b\}$ and $Z = \{5, 6, 7\}$. Consider the functions $f = \{(1, a), (2, a), (3, b)\}$ and $g = \{(a, 5), (b, 7)\}$. Find $g \circ f$.

(c) Define fundamental circuit matrix and generate it with respect to the given shaded spanning tree for the following graph.



(d) Explain the applications of graphs.

(e) Define the following with one example each :

(i) Hamiltonian circuit.
(ii) Planar Graph.

(f) Let (I, +) be a group, where I is the set of all integers and + is an addition operation. $Let\ H = \{\ldots. -4, -2, 0, 2, 4, 6, 8, \ldots..\}$ be the subgroup consisting of multiples of 2. Determine all the left cosets of H in I.

(g) Explain any 4 properties of binary relations.

(h) Consider a semi-group (A, *). Show that for a, b, c in A, if $a * c = c * a$ and $b * c = c * b$, then $(a * b) * c = c * (a * b)$.

(i) Explain with an example, how matrix multiplication helps to simplify a 2-layer cascaded system.

(j) Explain the roles of inner product and outer product in a PDP system.

(P.T.O.)

PART B

(4 × 10 = 40)

II. (a) Explain the principle of Mathematical Induction. Hence show that : (5)

$$1^3 + 2^3 + 3^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2 \ \forall n \in N.$$

(b) Consider the set $A = \{4, 5, 6, 7\}$ and the relation R on A given by (5)
$R = \{(4,5), (5,6), (5,7), (6,6), (6,7), (7,6), (7,7)\}$.
Determine (i) R^3 (ii) R^∞.

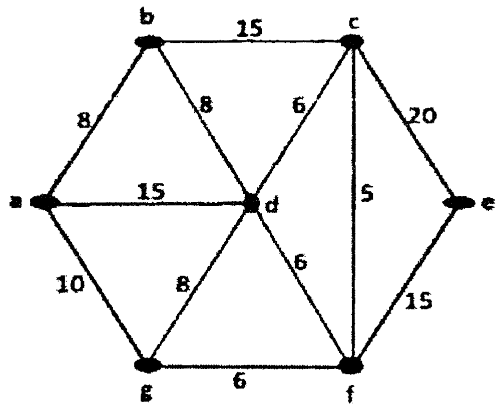
OR

III. (a) Consider the relation (5)
$R = \{(1,1), (1,3), (1,5), (2,2), (2,4), (3,1), (3,3), (3,5), (4,2), (4,4), (5,1), (5,3), (5,5)\}$
on the set $A = \{1, 2, 3, 4, 5\}$. Determine whether the given relation is an equivalence relation. If so, list its equivalence classes.

(b) Find the product $f \circ g$ and $g \circ g$ for the following permutations: (5)

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 3 & 1 & 2 \end{pmatrix}, \ g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 4 & 2 & 1 & 3 \end{pmatrix}.$$

IV. (a) Explain Prim's algorithm. Find minimal spanning tree of the following graph using Prim's algorithm : (7)

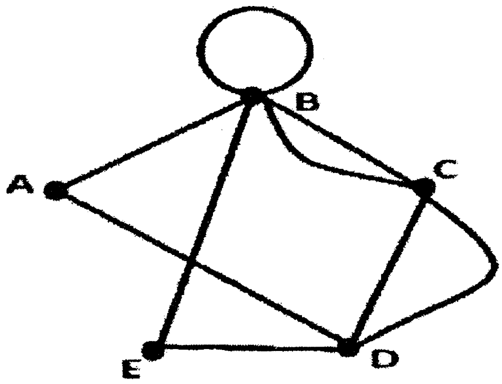


(b) Describe the characteristics of a Binary Search Tree. Explain how the following operations are done in a Binary Search Tree. (3)
   (i) Inserting elements in the order 55, 77, 22, 33, 66, 60, 63, 75 and 99 into an initially empty BST.
   (ii) Deleting elements 77 and then 99.

OR

(Contd...3)

V. (a) Explain the criterion for existence of Euler path and Euler circuit in a graph. Use Fleury's algorithm to construct an Euler circuit for the following gaph: (7)



(b) Compare isomorphism and 1- isomorphism. (3)

VI. (a) Consider an algebraic system (N, +) where $N = \{0,1,2,3,4,\ldots\}$ is the set of natural numbers and + is an addition operation. Determine whether $(N, +)$ *is a monoid.* (5)

(b) Show that identity element and inverse element in a group is unique. (5)

OR

VII. (a) Consider an algebraic system $(G,*)$ where G is set of all non-zero real numbers and * is a binary operation defined by $a * b = \frac{ab}{8}$. Show that $(G,*)$ is an abelian group. (5)

(b) Write short note on the following with examples : (5)
   (i) Rings.
   (ii) Symmetric groups.

VIII. (a) What are the characteristics of inner product? Explain the geometrical interpretation of inner product. (5)

(b) Write short notes on the following : (5)
   (i) Change of basis.
   (ii) Multilayer systems.

OR

IX. (a) Explain in detail Linear combination and Linear independence. Determine whether the following set of vectors is linearly dependent or independent: $S = \{v_1 = (1,2,3), v_2 = (0,1,2), v_3 = (-2,0,1)\}$. (5)

(b) Discuss the significance of Eigen values and Eigen vectors in vector space. (5)

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